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Are Women More Risk Averse? A Sequel

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Abstract: This paper reexamines the question of gender differences in financial relative risk aversion using updated methods and data. Specifically, the paper revisits the 1998 work “Are women more risk averse?” by Jianakoplos and Bernasek, suggests refinements in their model in relation to the database used, namely the U.S. Federal Reserve Board’s Survey of Consumer Finances (SCF), and performs new tests on the latest SCF from 2022. The suggested refinements pertain first to an enhanced computation of wealth, which includes additional categories of assets such as 401(k)s or other thrift savings accounts, and second to the more subtle handling and consideration of specific demographic data of the SCF respondents. Unlike the original study, which also included married couples, the new study focuses exclusively on single-headed (never-married) households. This eliminates ambiguity about the actual financial decision maker in households, enabling a clearer assessment of individual gendered behavior. Following the refinements, the new tests reveal a continuing pattern of decreasing relative risk aversion; however, contrary to the 1998 findings, there is no significant gender difference in financial relative risk aversion in 2022. This study also documents that education levels strongly influence risk-taking: single women with higher education levels are more likely to hold risky assets, while for men, higher education correlates with less risk-taking. The paper concludes by informing policymakers and financial educators so as to further tailor their strategies for promoting gender equality in financial decision-making.

Keywords: financial relative risk aversion; gender; gender equality; individual investors; survey of consumer finances; risk-free assets; risky assets; single-person households; household finances; financial decision-making



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1. Introduction

This paper reexamines the question of gender differences in financial relative risk aversion by using updated methods and data. Specifically, the paper tests whether women are more financially risk averse than men, revisiting the seminal 1998 work “Are women more risk averse?” by Jianakoplos and Bernasek. The current paper attempts an enhancement of the original [Jianakoplos and Bernasek \(1998\)](#) model followed by a subsequent series of new tests in the latest U.S. Federal Reserve Board’s Survey of Consumer Finances (SCF) of 2022. The suggested enhancements refer to a refined computation of wealth, as well as to a more subtle handling of a set of demographics of the SCF participants. The refinements to the definition of wealth pertain to the inclusion of additional categories of assets such as 401(k)s, which are deemed to help study a more nuanced portrayal of the financial risk-taking of the SCF households. Moreover, the refinements to the demographic

data of the SCF respondents secure the consistency of the econometric model. Lastly, unlike the original study, which also included married couples, this paper focuses exclusively on single-headed (never-married) households. This eliminates ambiguity about the actual financial decision maker in households, enabling a clearer assessment of individual gendered behavior.

For the 2022 SCF, the authors find that both single women and single men exhibit decreasing relative risk aversion as wealth increases; however, there is no significant gender difference in this decrease. In hypothetical investment scenarios, single men show a higher inclination for “above-average risk” compared to women, who more frequently choose “no risk”. Education levels strongly influence risk-taking: single women with higher education levels are more likely to hold risky assets, while for men, higher education correlates with less risk-taking. Finally, and contrary to the 1998 findings, there is no significant gender difference recorded in 2022.

In the authors’ humble opinion, the aforementioned finding of decreasing financial relative risk aversion for 2022, for both never-married female and male individuals, as well as the recorded finding of a lack of gender differences, represent two very interesting results. First, it is of interest that both women and men are recorded as keeping the pattern of decreasing relative risk aversion over time, since more financial risk-taking could potentially safeguard better returns on their investment portfolios. Second, the elimination of any gender gap in financial risk aversion, even if it is specific only for this cross-section, is an encouraging result for both women and men, as well as for financial educators and advisers, who have been working towards achieving gender equality in financial risk-taking. Lastly, the recorded lack of gender differences challenges the notion that women are inherently more risk averse than men in financial decisions. This result could be added to documented evidence against the standard conclusion in economics and finance that “women are more risk averse than men”, a conclusion that merits investigation, as [Nelson \(2015\)](#) suggests.

The paper is organized as following: Section 2 begins with a literature review of studies related to gender differences in financial risk aversion and financial risk tolerance, with a focus on more recent findings. Next, the econometric model that serves as the canvas of the paper, namely the [Jianakoplos and Bernasek \(1998\)](#) model, is roughly delineated, followed by the presentation of the current paper’s refined model along with coverage of the 2022 SCF dataset. In Section 3, the empirical results and the new findings of the paper are presented. Section 4 concludes with a policy discussion highlighting the importance of financial education and advice in the elimination of any gender gap in financial decision-making. Limitations of the research, as well as possible future directions, are also discussed.

2. Materials and Methods

2.1. Gender and Financial Risk Aversion in the Literature

Here, the authors wish to offer readers a glimpse into seminal, as well as equally important, more recent findings regarding gender differences in financial risk aversion, and the related areas of financial risk tolerance and investment behavior in general. Before anything else, it is stressed that when considering the effect of gender on risk aversion, as the review reports below, in many instances in the literature the basic premise is that women are more risk averse than men. This is a conclusion reached in several research settings; the settings may range from an analysis of traditional surveys that elicit risk aversion patterns, as [Embrey and Fox \(1997\)](#) show in their literature review, to new more elaborate methods that combine measurements of risk aversion with real-world investments brought into laboratory settings (e.g., in [Meziani and Noma \(2018\)](#)). Nonetheless, this also remains a debatable statement, and the pursuit of determining individual gendered behavior in

financial risk-taking could be called ongoing. In this brief coverage, the intention is to go beyond the premise that women are more risk averse than men in their financial decision-making and attempt to offer insights into the factors contributing to or eliminating possible gender differences.

The first reference is to the research article “Are women more risk averse?” by [Jianakoplos and Bernasek \(1998\)](#), which also serves as the basis for this paper. In this seminal and highly cited work (the work has more than 2822 citations), [Jianakoplos and Bernasek \(1998\)](#) reported finding that holdings in risky assets increased with wealth for all types of households, which meant that relative risk aversion followed a decreasing pattern; however, relative risk aversion did not decrease as much for single women as for single men and married couples, indicating that single women were relatively more risk averse.

Next, regarding recorded explanations of gender differences in financial risk-taking in the literature, [Sung and Hanna \(1996\)](#) documented that differences in financial risk tolerance by gender could be due to differences in the understanding of the nature of risk. [Sapienza et al. \(2009\)](#) even accounted for variation in salivary concentrations of testosterone and in markers of prenatal testosterone exposure to investigate between- and within-gender variation in financial risk aversion. The authors documented that higher levels of circulating testosterone were associated with lower risk aversion among women, but not among men; however, at comparably low concentrations of salivary testosterone, the gender difference in risk aversion disappeared, suggesting that testosterone had nonlinear effects on risk aversion regardless of gender.

[Cupples et al. \(2013\)](#) recorded empirical evidence that attained education mediated the association between gender and financial risk tolerance. First, the authors determined that the direct association between female and risk tolerance was negative. That is, women exhibited a risk averse profile; however, they further determined that the total effect of gender on risk tolerance could be significantly reduced when education was used as a mediator between gender and risk tolerance.

[Yao et al. \(2011, p. 885\)](#) reported empirical evidence that both married and unmarried females were less risk tolerant than married males at all risk levels; however, the authors also decomposed the effect of age on risk tolerance, concluding with a statistically significant effect of aging and period effects on financial risk tolerance.

[Fisher and Yao \(2017\)](#) decomposed gender differences in financial risk tolerance and provided empirical evidence that gender differences might be explained by gender differences in the individual determinants of financial risk tolerance. Specifically, according to the authors, the individual variables that moderated the relationship between gender and high risk tolerance were income uncertainty and net worth, with income uncertainty moderating the relationship between gender and some risk tolerance.

Other researchers in the literature approached gender differences through the prism of the theory of expectations. For instance, [Qadan and Ben-Zion \(2014, p. 41\)](#) reported empirical evidence that there was a significant difference between the expectations between males and females regarding movements of short-term and long-term economic and financial variables. More specifically, females tended to expect higher changes, while males tended to lower their expectations regarding future movements.

[Jacobsen et al. \(2014\)](#) found that women tended to be significantly less optimistic than men regarding a broad range of issues, including the economy and financial markets. When the authors took differences in optimism into account, systematic gender differences in asset allocations disappeared.

[Dawson \(2023, p. 940\)](#) focused on the psychological underpinnings of risk and reported that women were “found to report a lower willingness to take risks than men with 53% of

this gap explainable by the higher levels of loss aversion amongst women and a further 3% attributable to lower levels of financial optimism amongst women”.

In their own approach, [Bouchouicha et al. \(2019\)](#) illustrated how subtle differences in the definition of loss aversion could result in the estimation of contradictory gender effects. The authors used four definitions of loss aversion commonly used in the literature, and they showed that according to one definition women were more loss averse than men, whereas according to another there was no gender effect; using the remaining two, they found women to be less loss averse than men.

2.2. The [Jianakoplos and Bernasek \(1998\)](#) Econometric Model

The current paper used the econometric model suggested by [Jianakoplos and Bernasek \(1998\)](#) (JB from now on) as its basis. JB based their work on the theoretical framework and the measure of relative risk aversion of [Friend and Blume \(1975\)](#), and by using data from the 1989 SCF posited for any investor, k , the model for which is shown in Equation (1):

$$\begin{aligned} \text{ratio}_k = & \beta_0 + \beta_1 \ln(\text{wealth})_k + \beta_2 \text{race}_k + \beta_3 \text{kids}_k + \beta_4 \text{home}_k + \beta_5 \text{human}_k + \sum_{i=1}^{i=10} \gamma_i \text{age}_{ik} + \sum_{j=1}^{j=5} \delta_j \text{work}_{jk} \\ & + \sum_{m=1}^{m=5} \lambda_m \text{educ}_{mk} + u_k \end{aligned} \quad (1)$$

where ratio was the ratio of risky assets of the SCF household to wealth of the household; wealth was the liquid net worth of the household, defined as the sum of the net value of risky and risk-free assets of the household. [Jianakoplos and Bernasek \(1998, p. 623\)](#) classified “dollar balances in checking accounts, savings, and money market accounts, certificates of deposit, U.S. savings bonds, Individual Retirement Accounts (IRAs) balances invested in certificates of deposit or bank accounts, and the cash value of life insurance less policy loans outstanding” as SCF risk-free assets. Further, [Jianakoplos and Bernasek \(1998, p. 623\)](#) classified “the sum of balances in IRAs not invested in bank deposits, stock holdings less margin loans outstanding, bonds, trust assets, the net value of real estate owned other than residential housing, the net value of business owned, and the net value of other miscellaneous assets (e.g., precious metal, futures contracts, artwork) reported by the household” as SCF risky assets.

Race was a dummy variable equal to one if the SCF respondent was black and zero otherwise; kids was the number of people 18 years or younger in the SCF household; home was a dummy variable indicating whether the SCF respondent was a homeowner or not; and human was the ratio of human capital to wealth. Human capital was estimated by assuming that the wage, salary, and self-employment earnings of the household for the SCF year of consideration continue until retirement. In terms of retirement, JB adhered to the [Friend and Blume \(1975, p. 907\)](#) algorithm of computation.

Age was a set of dummy variables indicating into which of ten age categories the SCF household head fell. The constructed age categories were less than 25, 26–30, 31–35, 36–40, 41–45, 46–50, 51–55, 56–60, 61–65, and over 65. Work was a set of dummy variables indicating five categories of labor force status of the SCF household head. According to the JB classifications, the SCF household head could be categorized as self-employed, as employed by others, as retired, as a farmer, or, finally, as unemployed or not in the labor force. Lastly, educ was a set of dummy variables indicating which of five levels of educational attainment the SCF household head has reached, namely grade school or less, some high school, high school degree, some college, and college graduate.

2.3. The Refined Econometric Model

One of this paper’s major enhancements in the JB model lies in the use of a refined definition of wealth. This was driven by the study of the SCF, and the remark that certain SCF

asset categories have been omitted from the JB computations or have not been utilized to the fullest. For instance, this paper's study revealed that JB had not included any retirement accounts in their computations, although the survey allowed for such computations, as well as a further categorization of such accounts into risk-free and risky ones. In a later work from [Bajtelsmit et al. \(1999\)](#), Bajtelsmit, Bernasek, and Jianakoplos, once again using data from the 1989 SCF, focused entirely on defined-contribution pension balances (such as 403(b)s, etc.) to study gender differences; however, in the JB work that is studied here, such balances were not included.

On the contrary, researchers such as [Chang et al. \(2018\)](#) considered thrift savings plans in similar wealth calculations, and further proceeded with their own split of such plans into risk-free and risky ones. The current paper consulted the [Chang et al. \(2018, pp. 256–60\)](#) approach, as well as that of [Yuh and Hanna \(1997, p. 8\)](#), and also sought further advice from the algorithms of the SCF itself ([FED 1992](#)). As a result, the paper updated the JB split of SCF assets into risk-free and risky ones, and continued with an enhanced definition of wealth called $\text{wealth}_{\text{GK}}$. All computations and code are available upon request.

Next, in contrast to the JB approach, where the tests were implemented for single women, single men, and married couples, the current paper's focus shifted entirely to single-headed households. This decision was made firstly because of the inability to specify the gender of the true investment decision maker in the SCF, especially in the case of mixed-sex couples, and secondly for the sake of consistency in the econometric model.

Regarding the gender of the true investment decision maker, it is stressed that given the generic format of the only SCF investment question, "How is the money in (this/these) account(s) invested? Is it all in stocks, all in interest-earning assets, is it split between these, or something else?", assuming that the respondent is also the financial decision maker seems more natural in the single-headed households, but things become more complicated in the case of couples, either married or living together as partners. It is impossible to know if the two spouses or partners take joint investment decisions for their assets or if it is one of the two, perhaps again the respondent of the survey, who overbears in financial decision-making. JB themselves ([Jianakoplos and Bernasek 1998, p. 624](#)) acknowledged this difficulty, as did other researchers, e.g., [Embrey and Fox \(1997, p. 35\)](#), and concluded that "the most direct test of gender differences in portfolio allocation is between households headed by never married females, and households headed by never married males".

Regarding the consistency of the econometric model, particularly in earlier versions of the SCF, one more reason leading to work with single-headed households, linked to Equation (1), is that race is known only if the respondent is the head of the household ([Lindamood et al. 2007, p. 198](#)); however, the head is not always the respondent of the survey. One reason for this is suggested by [Lindamood et al. \(2007, p. 198\)](#), who mention that "because of the detailed nature of the financial data the SCF gathers, interviewers are trained to establish who is the most financially knowledgeable person in the household, and to interview that person", so this person may not be the designated head.

Thus, the current paper limits the comparisons between single females and males to ensure that all regressors in (1) refer to the same person. Regarding race, this paper also enhanced the definition, including not only black and non-black respondents but also white, hispanic, Asian, and other non-Asian or multiple races. A little tweaking was performed for educ too, considering respondents with less than high school, a high school diploma, some college, college graduates, a master's degree, and advanced degree holders.

A last minor adjustment of the paper refers to the variable work. It is mentioned that for this variable, farmers are left out from the categories considered. This was decided because a farmer is already categorized as self-employed or employed by others; thus, this additional classification was considered unnecessary. One more nuanced thought

here would be to proceed with a more refined classification, considering farmers who are self-employed versus self-employed individuals who are non-agricultural, and similarly farmers who are employed by others versus non-agricultural individuals employed by others. Nonetheless, the overall number of farmers in the sample was already limited. Thus, the paper proceeded assuming that self-employed farmers behaved in the same way as non-agricultural self-employed ones, while farmers employed by others behaved in the same way as individuals employed by any non-agricultural employer.

2.4. The Survey of Consumer Finances

For the purposes of this paper, the Survey of Consumer Finances (SCF) and its latest public dataset of 2022 (SCF 2023) were employed. The SCF is a triennial interview survey of a nationally representative sample of U.S. families, sponsored by the Board of Governors of the Federal Reserve System with the cooperation of the U.S. Department of the Treasury. The SCF aims at providing detailed information on the financial characteristics of U.S. households. The survey collects data on families' assets and liabilities, their current and past employment, their pensions, their income, and their consumer attitudes. Data on the demographics of the families are also collected. The SCF employs a dual-frame sample incorporating both an area probability (AP) sample and a special list sample developed from a sample of tax records and obtained under very strict rules governing confidentiality.

Since the SCF sample is not an equal probability design, it was mentioned beforehand that the descriptive statistics reported in the paper are sample-weighted. The weight used is given in variable $\times 42,001$. Additionally, and with regard to more complex statistics related to regressions, the paper uses sample weights again.

It is noted here that the use of weights in multivariate analyses has been the approach of choice in most works in the literature employing the SCF; however, it is reported that there are researchers, such as Lindamood et al. (2007), who characterized weighting in multivariate analyses as controversial. Deaton (1997, p. 66) commented that “weighting regression analyses when the weights are endogenous is suspect for hypothesis testing”.

Thus, the suggestion here would be to perform multivariate analyses both with and without the use of sample weights. This is to be found in Deaton (1997, p. 72), as well as in Lindamood et al. (2007, p. 212); however, in this paper, the authors do not make any suggestion about endogenous weights for the SCF. Thus, the paper uses sample weights in all multivariate analyses. Lastly, it is reported that the paper uses data from the first implicate of the five available implicates of the public dataset.

It is worth mentioning that JB, in their 1989 SCF analysis, treated not only those who never married but also widowed and divorced individuals as single people. Jianakoplos and Bernasek (1998, p. 624) acknowledged the issue of the “extent to which widows and/or divorced women, continue the investment strategies begun by their late or ex-husbands”, but still, they allowed for “biased results against finding any gender differences in financial risk-taking”. Nonetheless, the current paper, helped by a bigger sample size of never-married individuals in the 2022 SCF, and most importantly for the sake of the already-stated argument of pursuing the most direct test possible of gender differences, treated only the never-married as single people.

Moving on with the sample, it is reported that the 2022 SCF surveyed 4595 households, out of whom there were 748 heads of households or 16.28% identifying as never married. Following one more convention, also adopted by JB, the analysis includes single-person households with wealth exceeding \$1000. The wealth measure employed is this paper's refined $\text{wealth}_{\text{CK}}$. Table 1 reports the weighted descriptive statistics for single women and men in the 2022 SCF.

Table 1. Weighted descriptive statistics for single-headed (never-married) households in the 2022 SCF.

	Single Women	Single Men
ratio (ratio of risky assets to wealth _{GK})	0.36	0.43
human capital	882,444	1,426,613
human (ratio of human capital to wealth _{GK})	104.84	93.17
wealth _{GK}	179,695	283,105
age		
Less than or equal to 25	0.16	0.18
26–30	0.11	0.15
31–35	0.16	0.16
36–40	0.09	0.12
41–45	0.08	0.08
46–50	0.06	0.06
51–55	0.08	0.04
56–60	0.06	0.06
61–65	0.05	0.05
Over 65	0.13	0.10
work		
Self-employed	0.09	0.09
Employed by others	0.64	0.76
Retired	0.15	0.08
Unemployed/not in lf	0.12	0.07
educ		
Less than high school	0.02	0.04
High school diploma	0.14	0.23
Some college	0.33	0.26
College graduate	0.29	0.35
Master’s degree	0.19	0.09
Advanced degree	0.03	0.03
race		
White	0.55	0.67
Black	0.24	0.14
Hispanic	0.09	0.07
Asian	0.03	0.07
Other	0.09	0.05
kids	0.26	0.09
home	0.40	0.39
observations	264	256

Source: authors’ tabulations, Survey of Consumer Finances (SCF 2023).

In Table 1, we observe that for the 2022 SCF, single men appear as having higher wealth and human capital, as well as higher proportions of risky holdings than single women. Furthermore, 84% of single women have at least some college education, versus 73% of single men. Both single women and men are more likely to be employed by others,

while they are equally likely to be self-employed. The sample is mostly dominated by white followed by black individuals, while the proportions of the other races are lower. Lastly, single women are more likely than single men to have a kid or dependent in the household than single men, while both never-married female and male individuals are equally likely to be homeowners.

The Financial Risk-Taking Question in the 2022 SCF

Before moving to the empirical part that measures actual financial risk-taking, of possible interest in the SCF survey are the responses given to a hypothetical financial risk-taking question that the SCF uniquely offers. This SCF question is as follows: Which of the following statements comes closest to describing the amount of financial risk that you are willing to take when you save or make investments? The respondents may select one of the following: 1. Take substantial financial risks expecting to earn substantial returns, 2. Take above average financial risks expecting to earn above average returns, 3. Take average financial risks expecting to earn average returns, and 4. Not willing to take any financial risks. Table 2 isolates the weighted means for every response category and presents the results of weighted difference-in-means tests conducted to look for significance in the results.

Table 2. Percentages of responses to each of the 2022 SCF risk tolerance question categories and weighted difference-in-means tests between single women and single men.

Risk Tolerance	Single Women	Single Men
Substantial risk	0.04	0.05
Above-average risk	0.14	0.29 [†]
Average risk	0.46	0.47
No risk	0.36	0.19 [†]
Observations	264	256

Source: authors' computations, Survey of Consumer Finances (SCF 2023). [†] The difference is statistically significantly different than zero at 5% level or less.

By observing Table 2, it is remarked that there is no significant difference in the percentages of responses of single women and single men for the average as well as for the substantial risk categories. Nonetheless, it is observed that single women are significantly more present than single men in the no risk category, while single men are significantly more present than single women in the above-average risk-taking category. Thus, although the SCF survey question is a hypothetical question, we cannot avoid observing single women's higher predisposition than that of men towards selecting no risk as a response in their intended investments, and similarly single men's higher predisposition than that of single women towards selecting the above-average option.

3. Results

3.1. Empirical Findings Based on the Enhanced JB Model

This section presents the empirical findings following the estimation of the refined Equation (1). It is repeated that the enhanced JB model is based on an alternative wealth definition, $wealth_{GK}$, and the suggested tweaks in the demographic variables of the respondents. For the purposes of the data analysis, this paper worked with Stata/SE 17.0.

It is stressed that since the ratio in Equation (1) can only take values between zero and one, this paper used a maximum likelihood tobit regression procedure, which was first performed separately for never-married female and never-married male respondents. Next, and in accordance with [Jianakoplos and Bernasek \(1998, p. 624\)](#), Equation (1) was

estimated jointly for single women and single men, including a dummy variable female, which equaled one for single women and interacted with each explanatory variable in order to determine which coefficients were statistically different between single women and single men. Lastly, adjusted Wald tests that accounted for the use of sample weights were performed to examine whether the estimated equations between never-married women and never-married men were significant. Table 3 presents the results.

Table 3. Weighted tobit regression results in the 2022 SCF for wealth_{GK}.

	Single Women	Single Men
ln(wealth) _{GK}	0.124 *** (0.024)	0.103 *** (0.025)
26–30	0.072 (0.128)	−0.055 (0.099)
31–35	0.125 (0.136)	−0.204 * [†] (0.119)
36–40	0.114 (0.130)	0.017 (0.137)
41–45	0.234 * (0.157)	−0.087 [†] (0.142)
46–50	0.038 (0.151)	0.027 (0.125)
51–55	0.045 (0.152)	−0.081 (0.225)
56–60	0.274 ** (0.120)	−0.338 ** [†] (0.152)
61–65	0.238 (0.193)	−0.044 (0.148)
Over 65	−0.004 (0.220)	0.017 (0.208)
Employed by others	−0.298 ** (0.109)	0.058 [†] (0.130)
Retired	−0.215 (0.199)	−0.018 (0.209)
Unemployed/not in lf	−0.278 ** (0.151)	−0.021 (0.196)
High school diploma	0.767 *** (0.279)	−0.088 [†] (0.146)
Some college	0.798 *** (0.270)	−0.042 [†] (0.143)
College graduate	0.651 *** (0.278)	−0.202 [†] (0.148)
Master’s degree	0.638 *** (0.279)	−0.233 * [†] (0.146)
Advanced degree	0.798 *** (0.301)	−0.203 [†] (0.168)
Black	0.100 (0.076)	−0.107 [†] (0.088)
Hispanic	−0.083 (0.154)	−0.017 (0.170)
Asian	−0.052 (0.122)	−0.173 ** (0.076)
Other	0.064 (0.129)	−0.028 (0.152)

Table 3. Cont.

	Single Women	Single Men
kids	−0.007 (0.055)	−0.056 (0.085)
home	−0.093 (0.077)	0.024 (0.074)
human	−0.001 (0.001)	0.001 (0.001)
sigma hat	0.402	0.386
observations	264	256

Source: authors' computations, Survey of Consumer Finances (SCF 2023). Base: ≤ 25 , self-employed, less than high school, and white. *** p -value less than 0.01, ** p -value less than 0.05, and * p -value less than 0.10. † significantly different from the coefficient of single women at a 10% level or lower.

Before proceeding with the commentary on the 2022 SCF results for the refined definition of wealth, $\text{wealth}_{\text{GK}}$, an important note on the theory needs to be stressed. According to Friend and Blume (1975, p. 902), “the estimated coefficient on $\ln(\text{wealth})$ shows how the inverse of the coefficient of relative risk aversion, and thereby the coefficient of relative risk aversion varies with wealth. More specifically, a positive coefficient indicates decreasing relative risk aversion, while a negative coefficient indicates increasing relative risk aversion”.

3.2. New Findings in the 2022 SCF for $\text{Wealth}_{\text{GK}}$

In terms of the empirical findings, first, the Wald test adjusted for sample weights reveals that the estimated equations are significantly different between female and male single-person households ($F(26) = 40.55$ and $\text{Prob} > F = 0.0344$).

Next, one of the most prominent new findings of the paper has to do with the sign/magnitude of the estimated coefficient on $\ln(\text{wealth})_{\text{GK}}$. From the results in Table 3, it is deduced that, controlling for the effect of all other variables, both female and male never-married individuals show decreasing financial relative risk aversion. It is also observed that, contrary to the JB results for 1989, in the 2022 SCF financial relative risk aversion decreases slightly more for single women (coefficient of $\ln(\text{wealth})$ equal to 0.124) than for single men (coefficient equal to 0.103); that is, single female respondents are relatively less risk averse than single male ones.

Nonetheless, there is no significant difference between the single female and the single male $\ln(\text{wealth})_{\text{GK}}$ coefficient; that is, holding everything else fixed, as wealth increases single-headed households increase their risky holdings, showing decreasing relative risk aversion. However, there is no significant difference when women are compared to men. Thus, there are no gender differences recorded for risk aversion for 2022.

On another important note, and regarding the estimated coefficient of $\ln(\text{wealth})_{\text{GK}}$, it is worth mentioning that, given the tobit specification, and for the sake of econometric accuracy, the partial effect of $\ln(\text{wealth})_{\text{GK}}$ is not just $\hat{\beta}_1$ but $\hat{\beta}_1 \Phi\left(\frac{x\hat{\beta}}{\sigma}\right)$ (Greene 2003, p. 849, chap. 19). $\Phi\left(\frac{x\hat{\beta}}{\sigma}\right)$ is the tobit adjustment factor. In the formula, $\Phi()$ is the standard normal cumulative distribution function, x is shorthand for all of the explanatory variables, $\hat{\beta}$ is shorthand for all of the estimated coefficients, and σ is the standard deviation of the error term. Nevertheless, this adjustment factor, taking values between zero and one, is assumed not to change the results qualitatively. To investigate the latter idea, the following test was performed: considering the effect of all other variables, the tobit adjustment factor $\Phi\left(\frac{x\hat{\beta}}{\sigma}\right)$ was evaluated at the mean values of the corresponding samples, leading to the computation of an adjusted coefficient for $\ln(\text{wealth})_{\text{GK}}$. It is reported that the process gave qualitatively similar results to the ones of Table 3.

The commentary continues with the remaining regressors in the specification. First, we comment on the effect of age. From Table 3, it is observed that, regarding single women, with the exception of the over-65 age group, in all other age groups single women appear as taking higher risks in their portfolio than the omitted category of their less-than-or-equal-to-25 years old peers. This result is significant for the 41–45 and 56–60 age groups.

The picture changes for single men, in the sense that except for the 36–40, 46–50, and over-65 age groups, the other age groups are estimated to take lower financial risks than the base group of less-than-or-equal-to-25 years old peers. This result is significant for single men in the 31–35 and 56–60 age groups, but in the rest of the age groups it is noteworthy that the age coefficients are neither numerically high nor statistically significant. When we compare the two groups, the estimated coefficients between single females and single males are significantly different for the 31–35, 41–45, and 56–60 age groups.

Regarding the effect of work, the data analysis reveals that for single females, keeping everything else fixed, single females that are employed by others take less risk than the omitted group of self-employed females. This also applies to single women that are retired or unemployed or not in the labor force. When it comes to men, single males that are employed by others appear to take more financial risk than the base group of self-employed males. On the contrary, men categorized as retired or unemployed/not in the labor force are expected to take less risk than self-employed men, keeping all other factors constant. It is noteworthy that none of these effects for single men are either numerically high or significant; that is, work status seems to not affect single males' risky asset holdings. The estimated coefficients are significantly different only for the employed by others category when single women are compared versus single men.

Regarding the effect of education, a very interesting result arises from the observation of the relevant coefficients for single females. Keeping all other factors fixed, single females at all levels with more than a high school education are predicted to hold portfolios with much greater proportions of risky assets compared with those single women with less than a high school degree education. The result is significant for all education levels and suggests that, for single women, more years of education lead to significantly higher financial risk-taking.

The picture is different for single men, in that it is single men with less than a high school degree education who are estimated to hold higher percentages of risky assets in their portfolios than single men with more than a high school education. The estimated coefficients suggest that more years of education leave single men's financial risk-taking either unaffected or significantly decrease it, as in the case of college graduates. Lastly, for all levels of education, except for that of high school, single women are expected to have significantly higher risky holdings than single men.

Regarding the effect of race, keeping all factors fixed, black single females are estimated to take higher financial risks than white ones, but the effect is not significant. On the contrary, black single men are expected to take lower risk than white ones, but this effect is also not significant. Furthermore, the difference between black single females and black males is the only race-related difference, which is significant when comparing single females versus single males.

Continuing with the effect of race, hispanic single women and hispanic single men are estimated to have less risky holdings in their portfolios than their white peers, but both these effects are numerically and statistically insignificant. Next, Asian single women and Asian single men are expected to take less financial risk than their white peers, but this is a significant result only for Asian single men. Lastly, other or multiple-race single women hold fewer risky assets than their white peers, while other or multiple-race single men

hold more risky assets than their white peers; nonetheless, these are two numerically and statistically insignificant results.

Regarding kids, the interpretation of the estimated effect considers the following: With everything else fixed, if a household has one additional dependent, the ratio of risky assets held is expected to decrease for both single women and single men. However, the estimated effects are neither quantitatively nor statistically significant.

Regarding the effect of being a homeowner, the coefficients reveal that when an individual is a homeowner, financial risk-taking is expected to decrease for single women but is predicted to increase for single men. Nonetheless, similarly to the kid effects, these effects are quantitatively and significantly insignificant, and there is no significant difference in the coefficients when single women are compared to single men.

Finally, regarding human and its estimated effect, keeping everything else fixed, when the proportion of human capital relative to wealth increases, then the proportion of risky holdings for both female and male never married individuals is expected to remain the same. Thus, the quantitative effect is negligible. Moreover, the human regressor is insignificant for both groups.

4. Conclusions and Discussion

In this last section, the authors wish to highlight the significance and relevance of the work and provide some forward-looking conclusions. The focus is on the contemporary questions and practical implications of the paper's results as to what readers and policy makers alike could learn by reading the paper. Lastly, the possible limitations of the research are articulated, and future directions are sketched.

This paper aspired to bring to light new evidence about the effect of gender on the manifestation of financial relative risk aversion in investment decisions and investigate anew any gender differences. To this end, this paper employed a more comprehensive definition of wealth and performed a series of refinements on the model introduced in the seminal work of [Jianakoplos and Bernasek \(1998\)](#); it repeated a series of tests exploring gender differences in financial relative risk aversion with the help of data from the latest 2022 Survey of Consumer Finances (SCF). All tests were performed on weighted data, and focused on never-married households.

For the 2022 SCF, the paper found evidence of decreasing relative risk aversion; that is, as their wealth grew, both female and male single-person households showed decreasing financial relative risk aversion. However, there was no significant gender difference in this decrease. In the authors' humble opinion, this result updates an important line of inquiry, while it further challenges entrenched assumptions about gender and financial behavior. The findings contradict the traditional assumption in economics and finance that "women are more risk-averse than men"; thus, it could be claimed that the paper demonstrates how methodological enhancements and evolving societal contexts can reshape long-held conclusions in finance and economics.

One more important finding from the new study related to the role of education in women's financial risk-taking. In the empirical evidence, it was recorded that education levels strongly influence risk-taking for single women. This is a result on which policy-makers, as well as financial educators and advisors, could rely and use to further tailor their policies. This finding about education and its importance for single women could be linked with the recent finding by [Reiter-Gavish et al. \(2021\)](#) that women are more likely than men to utilize financial advice. [Brooks et al. \(2019\)](#) also observed that, following discussions with a financial advisor, the riskiness of the investment products selected by women were modified to a greater extent from their revealed risk preferences than those of men. Financial advice proves pivotal under distressed market conditions too; for instance,

[Reiter-Gavish et al. \(2022, p. 245\)](#) documented that women were more likely to leave the financial markets during the 2008 market crisis. However, financial advice reduced investors' tendency to leave the markets and had a positive correlation with their decisions to return to the market. Thus, women's tendency to abide to financial advice in combination with the positive effect of education on risk-taking could be used to the benefit of female investors to help them achieve higher returns on their long-term investments.

One further dimension to be emphasized here is the need not only for education, but for financial education and financial literacy in particular. [Hibbert et al. \(2013\)](#) recorded that, among highly educated individuals, women were more risk averse than men; however, they also recorded that when both men and women attained a high level of financial education, they were equally likely to invest a significant portion of their portfolio in risky assets, suggesting that financial education mitigated the gender difference in financial risk aversion. Moreover, [Bannier and Neubert \(2016\)](#) reported finding that, for women, standard investments were strongly associated with actual financial literacy. The authors suggested that actual financial knowledge appeared to be a prerequisite for women's risk tolerance. [Bannier and Neubert \(2016, p. 133\)](#) went a step further and suggested that raising women's actual literacy could reduce the gender gap in standard investments. Thus, overall, safeguarding access to both financial advice and financial education could better prepare female investors and eliminate any gap in financial risk-taking.

Before concluding, the authors wish to acknowledge certain limitations of the research. First, although the employed SCF wealth definition has been more comprehensive, there is always the concern as to what extent the SCF permits the successful deduction of risk-taking in the first place. Extracting financial information is always challenging, and the related SCF question about households' investment strategy is generic. One more limitation relates to the selected SCF subsample under consideration; that is, the subsample of single (never-married) individuals. The sample permitted a direct test of gender differences in financial relative risk aversion, comparing single women to single men; however, there is always the concern also raised by [Jianakoplos and Bernasek \(1998, p. 630\)](#) that there is no guarantee that the results could be generalized to all men and women.

Despite the aforementioned limitations, the authors deem that the refined JB model could be used in the future for more tests. A first suggestion could be to apply the model in a sequence of SCF cross-sections. This would significantly increase the size of the subsample of single-person households and could lead to statistically more valid inferences. Moreover, it would permit analysis over time. A second suggestion could be to apply the model to a panel of surveys from the SCF. This would allow for the study of gender differences in the same households. The above could be achieved thanks to the SCF panel element offered twice in the survey. In particular, the SCF re-interviewed the same households from 1983 in 1986 and 1989. Moreover, there is a panel element, with SCF households interviewed in 2007 being re-interviewed in 2009, at the time of the Great Recession, a period that could offer interesting grounds for further research.

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